

Enrique Valverde Sacristán

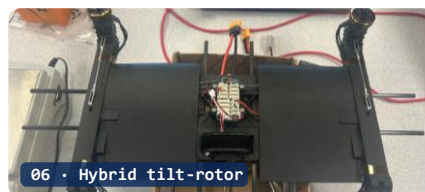
MSc Aerospace Engineering student, ISAE-SUPAERO (Aerospace Systems & Control)
Mechanical engineer

Model-based **state estimation** and UAV control, evaluated with **Monte Carlo campaigns** and **real flight data**. Looking to apply this to real robotic and mechanical systems.

LOOKING FOR

Mechanical engineering and robotics

Six-month final-year internship, from **March 2027**



03 INOAS: navigation with an intermittent GNSS receiver

UKF, pseudo-NIS sensor-fault alarm, NEES/NIS consistency, real GNSS error data · IEEE Aerospace 2027 (abstract accepted)

05 From simulation to flight: PID vs INDI

Real quadrotor flight logs, OptiTrack ground truth, error decomposition

06 INDI control of a hybrid tilt-rotor UAV

ENAC 6-DoF simulator, WLS allocation, tether sensitivity study

07 Sunspear: estimation for rocket airbrakes

IMU-driven Kalman filter, regime-scheduled sensors, MPC

08 Low-cost interceptor UAV

Camera/LiDAR mount, avionics, Betaflight; field test · team 1st place

09 Real-flight labs

Adaptive control with payload release; onboard sensing vs motion capture



CONTACT

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Toulouse, France · Spanish nationality

AT A GLANCE

MSc	ISAE-SUPAERO, Aerospace Systems & Control (2025–2027)
BSc	Mechanical Engineering, UC3M · Erasmus+ dual-degree programme at Arts et Métiers Paris
Industry	6-month internship, Safran Aircraft Engines
UAS	EASA Open A1/A3 and A2 remote-pilot certificates
Languages	English C1 (C1 Advanced, IELTS 7.5) · French B2 · Spanish native

WHERE EACH SKILL SHOWS UP IN MY WORK

Six projects at a glance

State estimation and flight control, backed by simulation and real flight-data analysis.

SKILL	INOAS 03-04	Flight test 05	Tilt-rotor 06	Sunspear 07	Interceptor 08	Labs 09
Real sensor and flight data logs, ground truth, error analysis	●	●				●
State estimation and sensor fusion Kalman filters, NEES / NIS, sensor scheduling	●	○	○	●		○
GPS-denied and degraded sensors GNSS off, biased sensors, IMU-only phases	●			●		
Sensor timing and latency sync filters, sensor lag, sample rates			●			●
Flight control INDI, PID, MRAC, MPC	○	●	●	○		●
Models and simulation 6-DoF simulators, Monte Carlo, sensor models	●	○	●	●		○
Hardware and flight tests integration, field and indoor flights		●			●	●

● core of the project ○ secondary

INDUSTRY

Safran Aircraft Engines, 2024. Six-month internship: finite-element models of a LEAP fan-case bond, kept in use by the design office after I left.

IEEE
Aerospace Conference 2027, 2nd author (abstract accepted)

250
paired Monte Carlo runs behind the INOAS results

10
real flights analysed against motion capture

1st
team place of four, low-cost drone challenge

WHAT I WOULD BRING
FEM experience at Safran, CAD and 3D-printed prototyping, plus control and estimation tested on real hardware, always checked against measured data.

TOOLBOX

KALMAN / UKF

ANSYS FEM

CAD: CATIA / SOLID EDGE

MPC

INDI

WLS ALLOCATION

PYTHON

MATLAB / SIMULINK

GIT

FLIGHT-LOG ANALYSIS

C / FREERTOS

BETAFLIGHT

SPACE GNC · STUDENT AEROSPACE CHALLENGE → IEEE AEROSPACE CONFERENCE 2027 (2ND AUTHOR, ABSTRACT ACCEPTED)

INOAS: navigating with a GNSS receiver that is mostly off

A LEO CubeSat that avoids debris on its own while its GNSS receiver sleeps **70 % of the time**.

PROBLEM

An always-on GNSS receiver drains a CubeSat. Switched off, the navigation error grows.

APPROACH

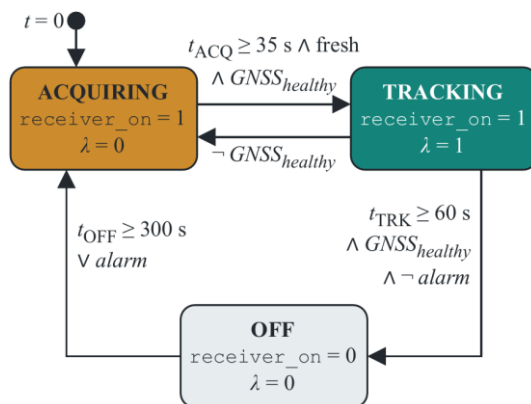
UKF + supervisor that wakes the receiver on a sensor-fault alarm + MPC whose keep-out grows with uncertainty.

RESULT

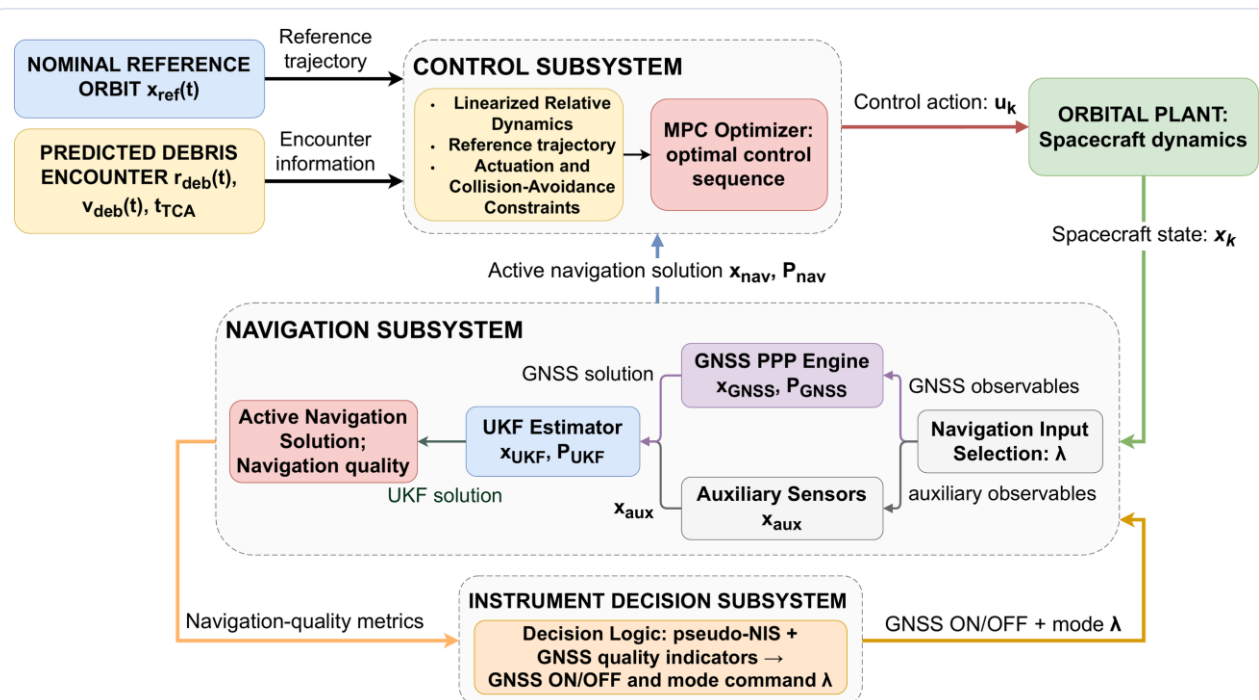
Median max error 8.0 m vs 21.1 m with a fixed timer; -66 % modelled receiver energy; no keep-out violation in 250 simulated runs.

MY PART

- UKF navigation, its process-noise budget and the three-state receiver supervisor.
- MPC guidance whose keep-out grows with the navigation covariance.
- Monte Carlo campaigns and all paper figures.



Receiver states. OFF → ACQUIRING → TRACKING; an alarm or a timer wakes it up.



Architecture. Navigation quality decides when to power the receiver; the navigation covariance sets the MPC keep-out radius.

INOAS · EVALUATION OVER 250 PAIRED MONTE CARLO RUNS

Judging a filter by its consistency, not only its error

Bounded error, less receiver energy, and a covariance that is **no longer overconfident**.

8.0 vs 21.1 m

median max position error, reactive vs fixed timer
(lower in 50/50 pairs)

3–7 s

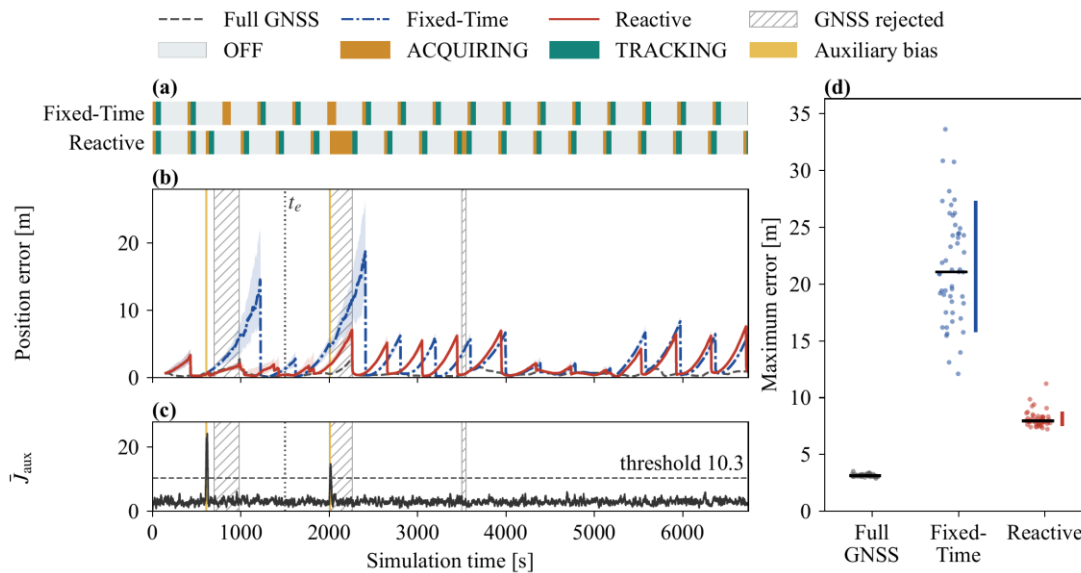
to flag an injected +5 km sensor bias

–32 %

Δv , covariance-adaptive vs constant 295 m
keep-out

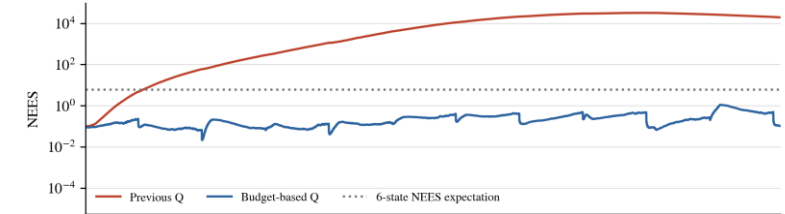
–66 %

modelled receiver energy vs always-on GNSS



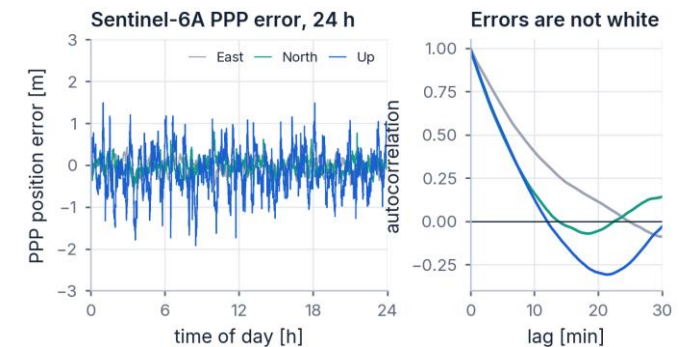
CONSISTENCY

Overconfident filter: NEES **12,650** → **0.23** (expected 6)



REAL SENSOR DATA

Sentinel-6A GNSS errors: 0.58 m RMS, but **correlated for minutes**



UAV FLIGHT DATA · ISAE-SUPAERO × ENAC RESEARCH PROJECT · 31 MARCH 2026

From simulation to flight: PID vs INDI

In flight, most of the error was **one offset shared by both controllers**.

QUESTION

Does INDI's advantage in simulation survive real flight data?

TEST

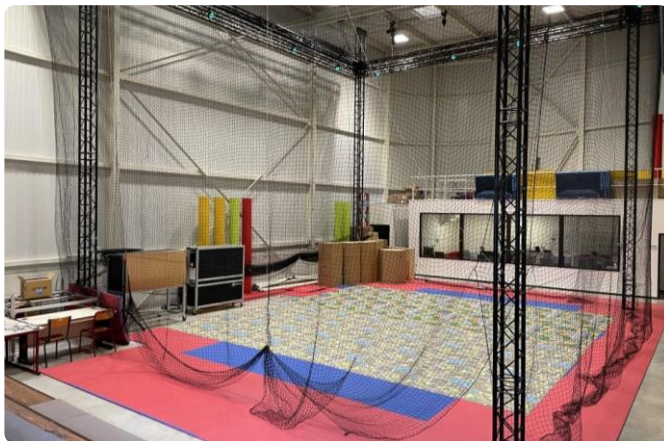
Same 2 m circle and metrics in simulation and in the ENAC arena, OptiTrack as ground truth.

FINDING

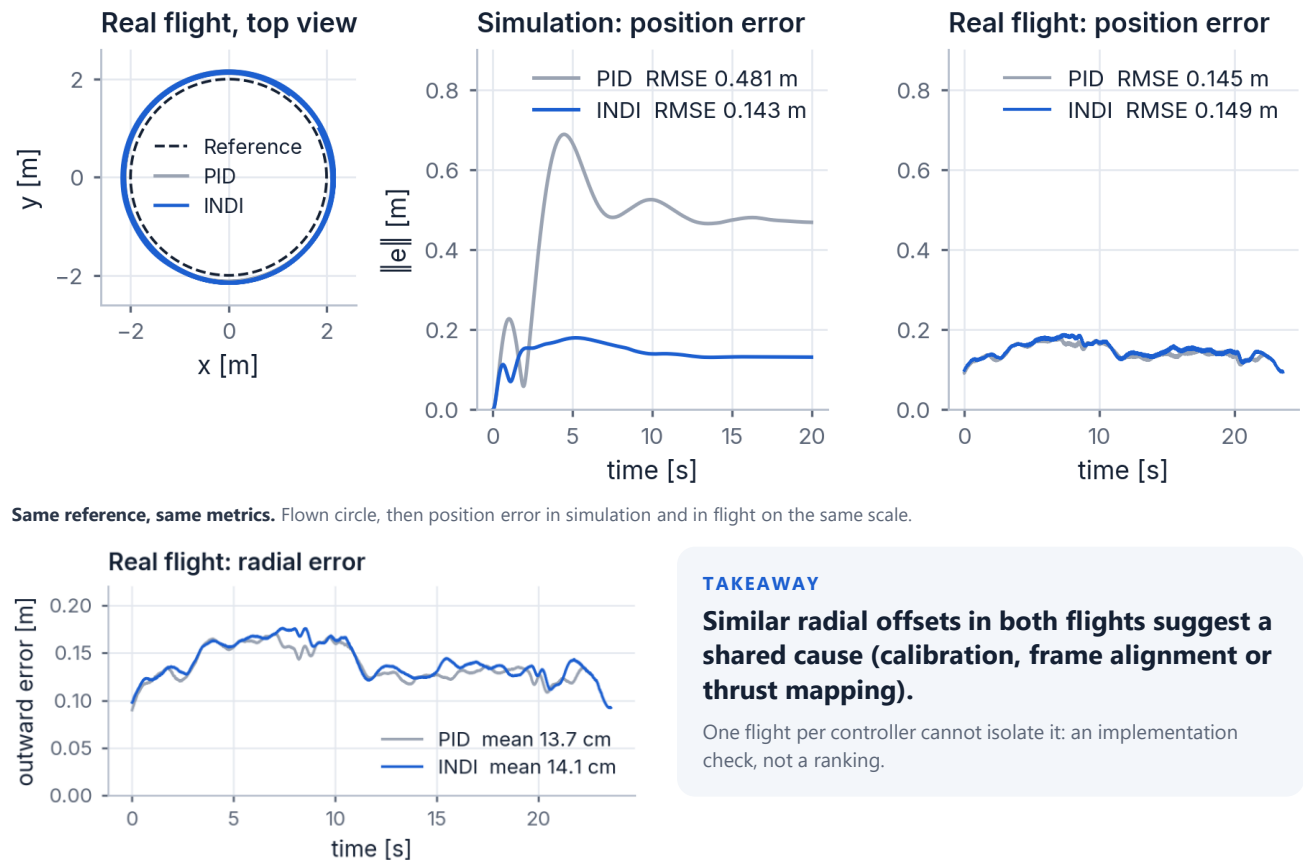
Sim: 70 % lower error with INDI. Flight: both ≈ 15 cm, mostly one shared 14 cm outward offset.

MY PART

- Co-wrote the baseline flown in the test (a PD + P cascade).
- Wrote the log-processing and metrics pipeline.
- Error decomposition that exposed the shared offset.



ENAC flight arena with OptiTrack cameras.



FLIGHT CONTROL & SIMULATION · ISAE-SUPAERO × ENAC RESEARCH PROJECT · SINCE FEB 2026

INDI control of a hybrid quad-tiltrotor, free and tethered

Accurate on the nominal orbit; a $\pm 15\%$ sweep showed **where it breaks**.

PROBLEM

Over-actuated tilt-rotor (4 rotors, 2 tilts, 1 wing) pulled by a tether that goes taut and slack.

APPROACH

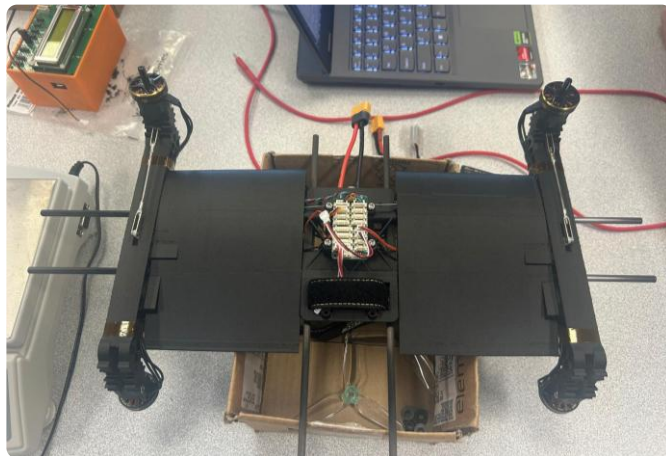
Cascaded INDI with WLS allocation on ENAC's 6-DoF simulator; one-at-a-time $\pm 15\%$ sweep.

RESULT

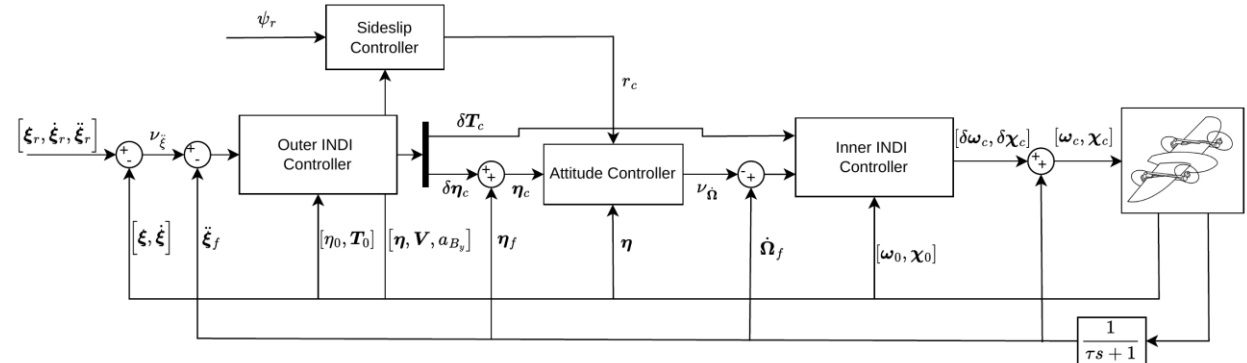
8.7 cm RMSE on the nominal tethered orbit; 8 of 16 perturbations diverge.

MY PART

- Found the INDI synchronisation filter effectively off ($\tau = 0$) in the simulator.
- Tether sensitivity campaign and its standalone report.
- Co-authored the 75-page report and defence.



Target platform, not yet flown with this controller.



Cascaded INDI. 100 Hz guidance and 200 Hz stabilisation loops, each allocating with constrained WLS.

	-15 %	+15 %
Cable stiffness k_c	diverges	diverges
Cable damping d_c	RMSE 11 cm	RMSE 8 cm
Cable length L_c	diverges	diverges
Linear density μ	diverges	diverges
Cable diameter	RMSE 11 cm	RMSE 8 cm
Segments N	RMSE 26 cm	diverges
UAV mass m	diverges	RMSE 10 cm
Orbit speed v_{max}	RMSE 6 cm	RMSE 44 cm

WHY IT FAILS

The orbit sits at the taut/slack boundary.

With almost no pre-tension, a perturbed UAV loses altitude, the cable goes slack, then snaps taut (198 N peaks in the stiffness case).

STATE ESTIMATION · SUPAERO SPACE SECTION · SINCE SEPTEMBER 2025

Sunspire: estimation for a supersonic rocket's airbrakes

Baro and GNSS become unreliable in flight, so the filter **picks its sensors by flight regime**.

PROBLEM

Airbrakes can only remove energy: they need a reliable altitude and speed early in the coast.

APPROACH

IMU-driven Kalman filter: accelerometer-bias state, density-scaled baro variance; linear MPC.

RESULT (SIMULATION)

Apogee error ≈ 12 m in 6-DOF co-simulation, with the MPC partly fed by the true state.

1 · BOOST, $M < 0.9$
IMU + barometer

2 · SUPERSONIC, > 4 g
IMU only: dead reckoning

3 · SUPERSONIC COAST
IMU + GNSS velocity

4 · SUBSONIC COAST
IMU + baro + GNSS · airbrakes active

launch

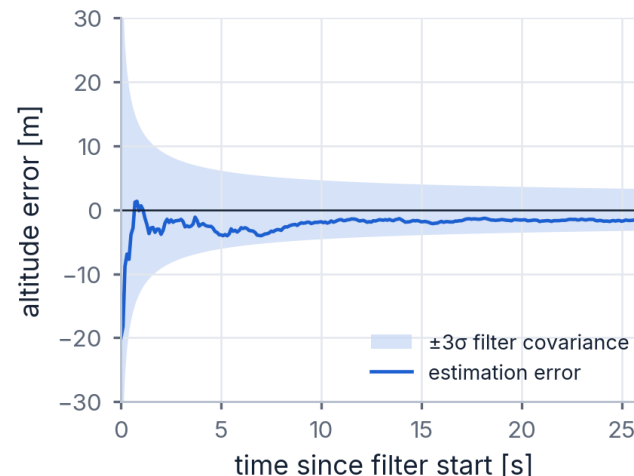
sensors selected by flight regime

apogee

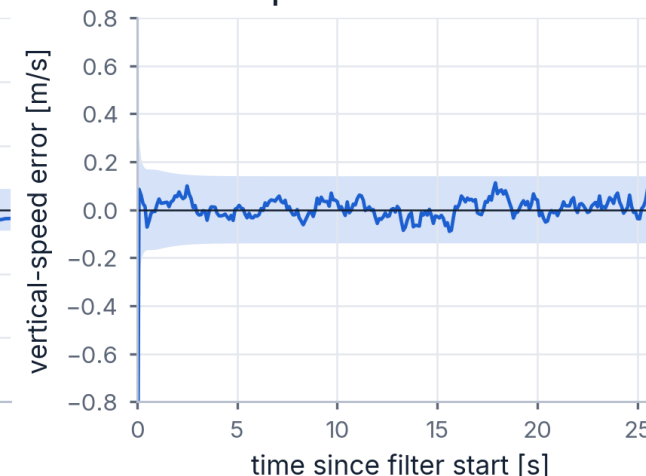
MY PART

- Co-developing the coasting-phase Kalman filter and the MPC chain.
- Per-step re-linearisation of the drag dynamics, now used by the MPC.
- Helping port the filter to C (FreeRTOS, ESP32-S3).

Altitude error



Vertical-speed error



Coasting-phase filter (Python re-run). From 20 m and 2 m/s off, the error stays inside its own $\pm 3\sigma$ bounds.



Sunspire CAD (structure team). Target apogee 9 km.

UAV HARDWARE & FIELD TEST · ISAE-SUPAERO LOW-COST DRONE CHALLENGE · DEC 2025 – JUN 2026

Low-cost interceptor UAV: from sketch to field test

Built with a team of six for the ISAE-SUPAERO drone challenge: **1st place of four**.

CHALLENGE

Two identical low-cost FPV drones,
≤ €3k per team.

VEHICLE

7-inch FPV quad, camera + ToF LiDAR in the nose,
no GNSS.

RESULT

Field-tested on 26 May 2026;
team 1st place.

MY PART

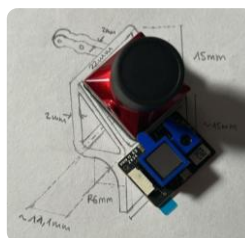
- Camera + LiDAR nose cradle: five CAD iterations in eight days, printed and flown.
- Avionics integration spec: FC soft-mount, LiDAR UART, EMI.
- Betaflight config, incl. LiDAR range on the OSD.



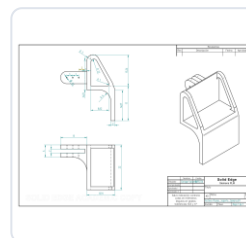
Field test.



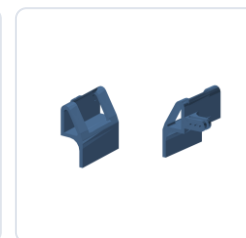
Prototype. Camera and LiDAR in the printed nose cradle.



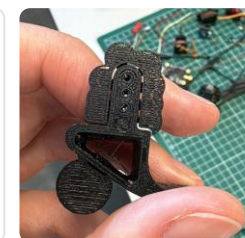
1 Sketch



2 Drawing



3 CAD v5



4 Printed



5 Flown

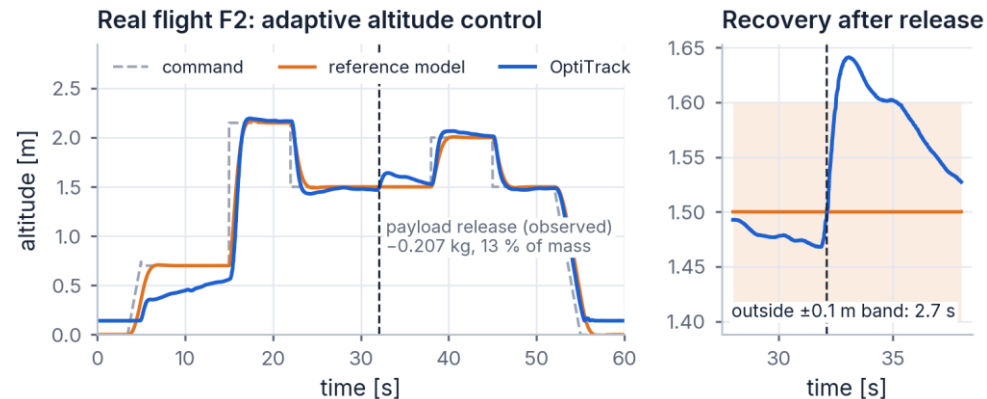
FLIGHT-TEST DATA · ISAE-SUPAERO MSC LABS · APRIL – JUNE 2026

Two flight-control labs on real quadrotors

Each controller went **simplified model** → **Simulink simulator** → **real flight**, scored against motion capture.

A · ADAPTIVE CONTROL

The drone drops 13 % of its mass mid-flight; MRAC adapts its altitude gains.



2.6–2.9 s

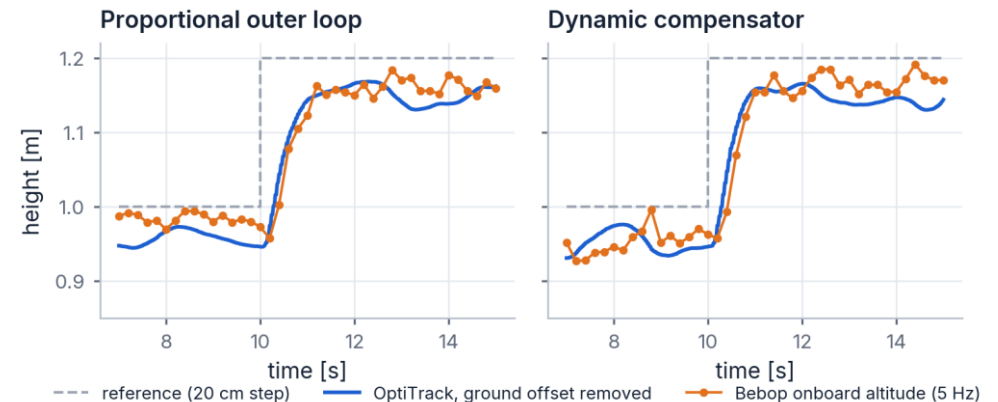
outside ±0.1 m of the reference model after the release, 5 flights

≈ 0.06 m

steady altitude RMSE on the plateaus

B · SENSOR LATENCY

Parrot Bebop via ROS: onboard altitude against motion capture.



≈ 0.2 s

lag of the onboard altitude (one 5 Hz sample)

5 cm

datum offset between the two altitude sources, corrected in the plot

AVAILABLE FROM MARCH 2027

Six-month final-year internship

ISAE-SUPAERO MSc, Aerospace Systems & Control
English C1 · French B2 · Spanish native
EASA A1/A3 + A2 remote-pilot certificates

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